Ambient transport and internal pair deficits

Human Compression project: L044 continuation

Let $G = \langle \rho_0, \dots, \rho_{r-1} \rangle$ be a transitive finite string action on Ω. Generators are involutions or identities. Fix $0 \le k < r$ with the ambient omission G_k intransitive, and put

$$P = G_{<k}, \qquad K = G_{>k}, \qquad H = G_{\ge k}.$$

Thus P centralizes K, and $G_k = PK$. Call an actual H-orbit *good* when K is transitive on it, and *bad* otherwise. These names are confined to this note; they express no source status.

Lemma 1 (A whole-orbit transport)**.** *For every good H-orbit O there are $p \in P$ and a bad H-orbit O' such that O^p is a K-orbit contained in O'. The bijection $O \longrightarrow O^p$ commutes with every marked generator ρ_j, $j > k$.*

Proof. The set O is one K-orbit. Since P centralizes K, its translates O^p are K-orbits, and their union B is the $G_k = PK$-orbit containing O. It is a proper subset of Ω. If every O^p were itself an H-orbit, ρ_k would preserve each one and hence preserve B. Then B would be invariant under both G_k and ρ_k, contrary to transitivity of G. Choose a translate $W = O^p$ which is not an H-orbit. Its containing H-orbit O' strictly contains W, so K is intransitive on O'. Commutation with all ρ_j, $j > k$, follows from $[P, K] = 1$. □

An edge of colour j is *ambient eligible* if its endpoints lie in different G_j-orbits, where G_j is the full ambient omission. Suppose a set of tail colours $J \subseteq \{k, \dots, r-1\}$ is assigned to actual H-orbits, with each colour assigned exactly once. For every assigned colour choose two ambient-eligible edges in its assigned orbit.

Theorem 2 (Simultaneous assignment normalization)**.** *The assigned pairs can be replaced by pairs of the same colours so that every colour is assigned to a bad H-orbit. Each new pair remains ambient eligible. In particular it is eligible for the full local omission on its new orbit.*

Proof. There is no eligible k-edge in a good orbit: $K \le G_k$ is already transitive there. Thus a colour j assigned to a good orbit satisfies $j > k$. Choose the element p from Lemma 1 for that orbit and transport both selected edges by p. They remain distinct actual j-edges, since p commutes with ρ_j. Also $P \le G_j$, so p preserves each G_j-orbit; the transported endpoints remain in different such orbits. Both edges lie in the same bad target orbit O'.

Do this for all colours assigned to good orbits, leaving every original bad-orbit assignment in place. A colour still has just one assignment, and no assigned pair is subsequently moved away from a bad target. Different colours cannot cause a conflict in this construction. Finally, the local omission on O' is the restriction of $\langle \rho_i : i \ge k,\ i \ne j \rangle \le G_j$. Distinct ambient omission orbits therefore imply distinct local omission orbits. This uses a subgroup inclusion, not equality of the two orbit partitions. □

Corollary 3 (The first-colour deficit)**.** *Suppose all tail indices are assigned, so $J = \{k, \dots, r-1\}$. Use the normalized assignments, and let F_a be the colours assigned to the nontrivial H-orbit O_a, of size m_a. Write $d_a = |F_a| - m_a/2$, and let p be the unique index with $k \in F_p$. Then*

$$d_p \le 0, \qquad d_a \le \begin{cases} -1, & m_a \text{ even}, \\ -\frac{1}{2}, & m_a \text{ odd} \end{cases} \quad (a \ne p). \tag{1}$$

An empty assignment has its exact deficit $d_a = -m_a/2$.

Proof. All full local two-edge hypotheses follow from Theorem 2. The accepted selected-colour theorem gives $d_p \le 0$. For $a \ne p$ with $F_a \ne \varnothing$, the orbit is bad and the selected subgroup lies in its intransitive full local omission of ρ_k. Hence $2|F_a| \le m_a - 1$. Integrality gives (1). If F_a is empty, its exact deficit implies the same bounds because $m_a \ge 2$. No transitivity of a selected or projected subgroup is being assumed. □

Theorem 4 (An internal centralizer and its exact boundary)**.** *Let O be an actual H-orbit of size $2q$, where $q \ge 3$. Suppose an involution τ centralizes H, preserves O and acts nontrivially on it. For a set $F \subseteq \{k, \dots, r-1\}$, suppose $\tau \in G_j$ and at least one ambient-eligible j-edge lies in O for every $j \in F$. Then*

$$|F| \le q - 1. \tag{2}$$

In equality, the pair images of the colours in F are exactly the transpositions of a path on the q pairs of τ in O. Their indices are consecutive; every other tail generator has trivial pair image.

Suppose additionally that $k \notin F$ and K is intransitive on O. In equality $K|_O$ is the diagonal S_q on two natural rows, and every tail mark outside $F \cup \{k\}$ fixes O pointwise. After row labelling, $\rho_k|_O$ flips either all pairs, just the first pair, or all pairs except the first. The last two alternatives require that the first path label is $k+1$.

Proof. An element centralizing a transitive group which fixes one point fixes every point. Thus $\tau|_O$ is fixed-point-free and has q pairs, preserved by H. Choose one ambient-eligible edge for each $j \in F$ and project it to the pair set. Its endpoints are distinct pairs because $\tau \in G_j$. The projected edges form a forest. Indeed an alternate projected path avoiding colour j lifts to a path using the other selected generators and τ, all in G_j, contradicting eligibility of the chosen j-edge. This also excludes repeated projected edges with distinct colours. The forest bound proves (2).

Suppose equality holds. The forest is a spanning tree T. For each $j \in F$, removing its edge gives two subtrees. The selected edges of each subtree, together with τ, connect all points above that subtree inside one G_j-orbit. The two subtrees lie in different G_j-orbits, by the chosen edge's eligibility. Consequently these are exactly the intersections with O of the two G_j-orbits which meet O.

Every other tail generator preserves O, its pairs, and both of these cut parts. A colour $i \in F$ preserves every tree cut except possibly its own, so its pair image fixes all vertices other than the endpoints of the i-edge. It exchanges those endpoints, hence is exactly that transposition. A tail colour outside F preserves every cut and fixes all pair vertices. The image of H on pairs is therefore the full S_q.

Incident edges of T give noncommuting transpositions, so their indices must differ by one. Three edges cannot meet one vertex, because three distinct integers cannot be pairwise consecutive. Thus the tree is a path. Its successive indices differ by one and cannot reverse direction without repeating an index. The indices are consecutive in one of the two path orientations.

Now suppose the additional hypotheses hold. Since K contains all marks in F, it is transitive on the pair set. Its assumed intransitivity on points gives exactly two orbits of size q, each meeting each pair once. A pair-trivial element of K fixes each row and hence every point. Thus $K|_O$ is faithfully the diagonal S_q, and the marks outside $F \cup \{k\}$ are identities there. Relabel the two rows accordingly.

The mark ρ_k has trivial pair image, so is a vector of pair flips. If $k+1 \notin F$, it commutes with all path transpositions and the vector is constant. If $k+1 \in F$, the path begins there; commutation with the remaining path fixes a common value on the other $q-1$ pairs, with an independent value on its first pair. The zero vector would leave H intransitive on O. The nonzero vectors are exactly those stated. □

For odd q, the three displayed flip vectors have respectively odd, odd and even parity on O. In particular the last local configuration cannot be discarded by parity alone. The theorem

proves deficit $|F| - |O|/2 \leq -1$ and describes equality; it does not assert the stronger deficit -2. The internal source case has odd $\tau|_O$, which gives q odd. A two-point orbit has no ambient-eligible edge of a colour whose omission contains τ, so its assigned set is empty and its deficit is exactly -1.

Source and exact boundary. The orbit-transfer strategy is borrowed from Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Proposition 5.27, physical page 36, and the first-colour deficit is used in Proposition 5.31 on page 42. The proof here specifies the transported whole K-orbit and the proper ambient G_k-orbit which forces a bad target. The internal pair forest and its equality proof supply a separate interface for the internally acting central involution in Proposition 5.31, cases (2)–(3), physical pages 42–43. They use only actual ambient omission cuts and the elementary tree argument, not a projected IP or an inferred classification of an arbitrary symmetric marking. The entire accepted selected-colour proof and its dependencies remain paid. No intersection property, parity or finite computation is used.

Crucially, the chosen pairs in Theorem 2 are *ambient* eligible. Two edges eligible only for a local tail omission have not been shown to remain eligible after transport by P. Nor is a pair split between two different H-orbits an input pair for this theorem. Applying the note requires the stated assignment to be supplied; it is not inferred from a bare local two-fracture condition. This does not establish the stronger internal odd-orbit deficits or their equality classification in source Proposition 5.31(2)–(3), and does not complete L044.